

From GPT-2 to gpt-oss: Analyzing the Architectural Advances

And How They Stack Up Against Qwen3



SEBASTIAN RASCHKA, PHD

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OpenAI just released their new open-weight LLMs this week: gpt-oss-120b and gpt-c20b, their first open-weight models since GPT-2 in 2019. And yes, thanks to some clever optimizations, they can run locally (but more about this later).

This is the first time since GPT-2 that OpenAI has shared a large, fully open-weight model. Earlier GPT models showed how the transformer architecture scales. The 2022 ChatGPT release then made these models mainstream by demonstrating concrete usefulness for writing and knowledge (and later coding) tasks. Now they have shared some long-awaited open-weight model, and the architecture has some interesting details.

I spent the past few days reading through the code and technical reports to summarize the most interesting details. (Just days after, OpenAI also announced GPT-5, which I will be discussing in the context of the gpt-oss models at the end of this article.)

Below is a quick preview of what the article covers. For easier navigation, I recommend the Table of Contents on the left of the article page.

- Model architecture comparisons with GPT-2
- MXFP4 optimization to fit gpt-oss models onto single GPUs
- Width versus depth trade-offs (gpt-oss vs Qwen3)
- Attention bias and sinks
- Benchmarks and comparisons with GPT-5

I hope you find it informative!

1. Model Architecture Overview

Before we discuss the architecture in more detail, let's start with an overview of the two models, gpt-oss-20b and gpt-oss-120b, shown in Figure 1 below.

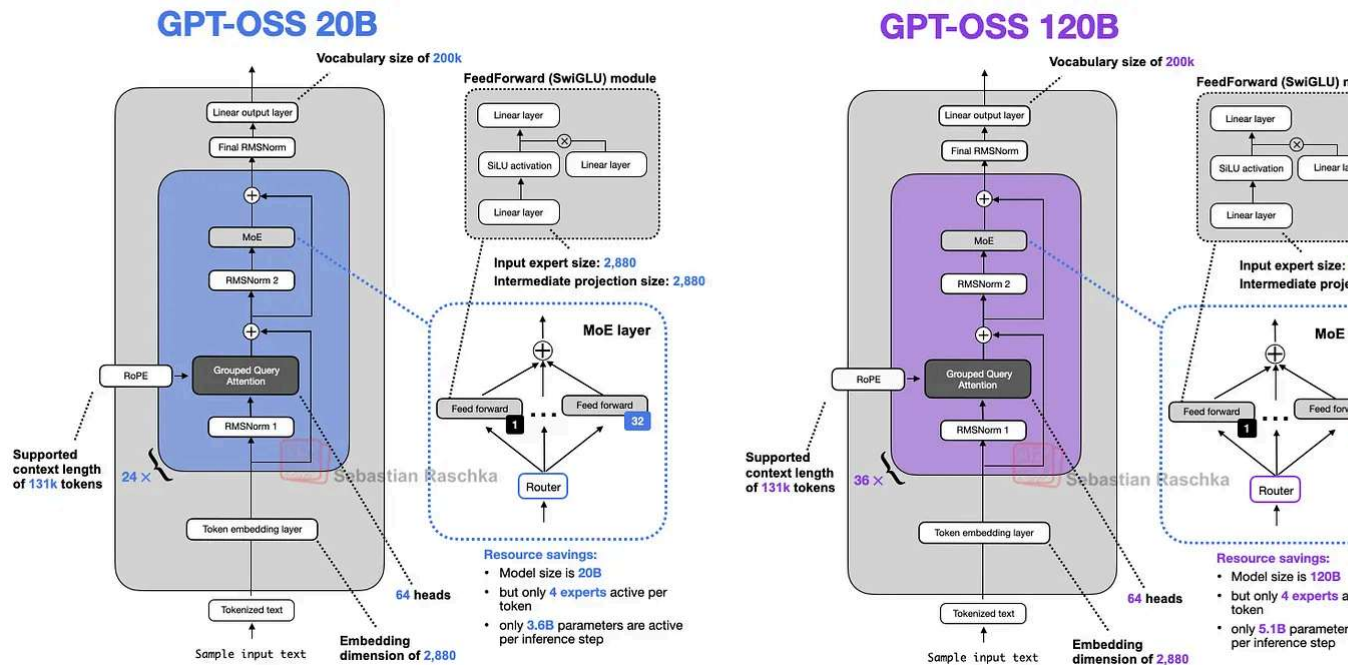
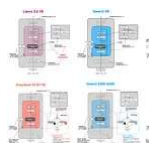


Figure 1: The two gpt-oss models side by side.

If you have looked at recent LLM architecture diagrams before, or read my previous [Big Architecture Comparison](#) article, you may notice that there is nothing novel or unusual in this architecture.



The Big LLM Architecture Comparison

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This is not surprising, since leading LLM developers tend to use the same base architecture and then apply smaller tweaks. This is pure speculation on my part, but I think this is because:

- There is significant rotation of employees between these labs.
- We still have not found anything better than the transformer architecture. Even though state space models and text diffusion models exist, as far as I know no one has shown that they perform as well as transformers at this scale. (Most of the comparisons I focus only on benchmark performance. It is still unclear how well the models handle long context or other tasks.)

real-world, multi-turn writing and coding tasks. At the time of writing, the highest-ranking non-purely-transformer-based model on the [LM Arena](#) is Jamba, which is transformer–state space model hybrid, at rank 96. EDIT: [Someone kindly pointed out](#) that there's a higher-ranking hybrid model: Hunyuan-TurboS at rank 22.)

- Most of the gains likely come from data and algorithm tweaks rather than from major architecture changes.

That being said, there are still many interesting aspects of their design choices. Some are shown in the figure above (while others are not, but we will discuss them later as well). For the rest of this article, I will highlight these features and compare them to other architectures at a time.

I should also note that I am not affiliated with OpenAI in any way. My information comes from reviewing the released model code and reading their technical reports. If you want to learn how to use these models locally, the best place to start is OpenAI's official model hub page.

- <https://huggingface.co/openai/gpt-oss-20b>
- <https://huggingface.co/openai/gpt-oss-120b>

The 20B model can run on a consumer GPU with up to 16 GB of RAM. The 120B model can run on a single H100 with 80 GB of RAM or newer hardware. I will return to this later, as there are some important caveats.

2. Coming From GPT-2

Before we jump into comparisons between gpt-oss and a more recent architecture, let's go back into the time machine and take a side-by-side look at GPT-2 (Figure 2) to see just how things have come.

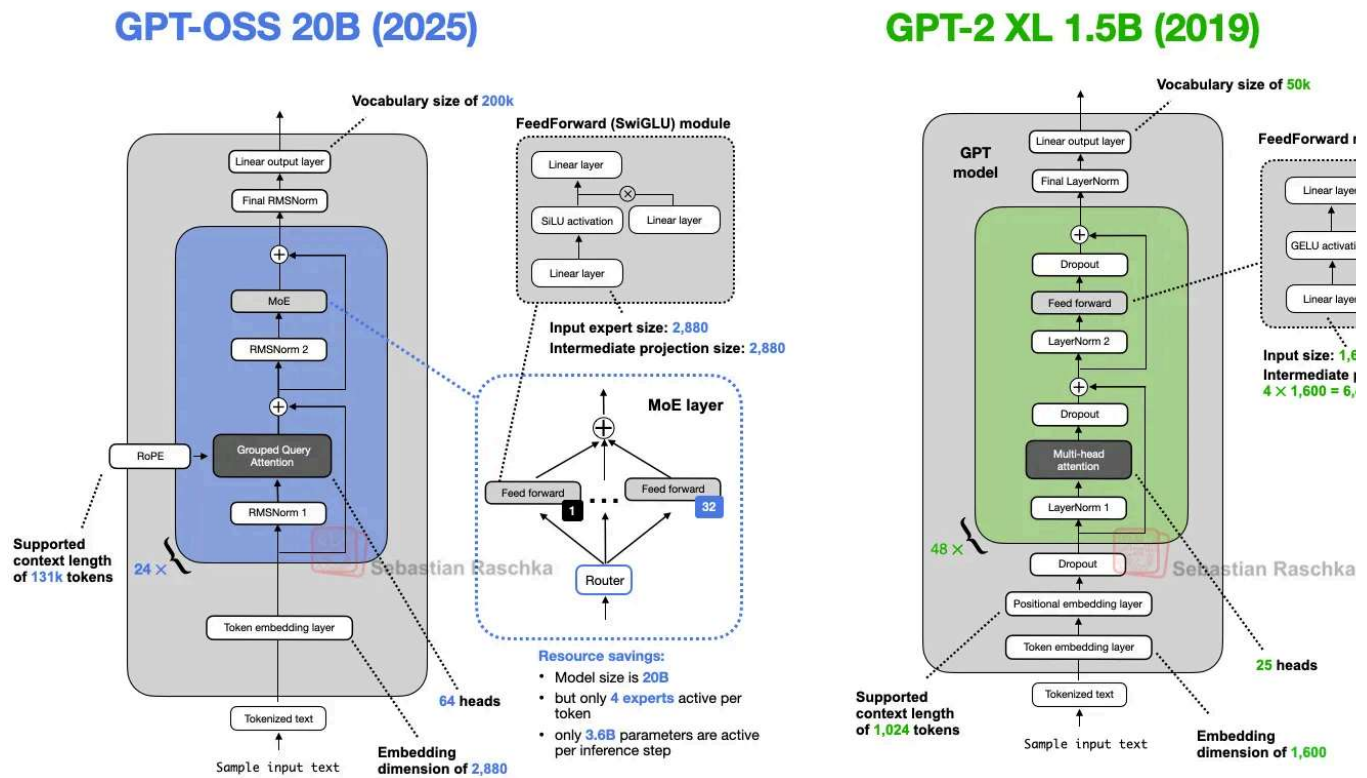


Figure 2: A side-by-side comparison between gpt-oss-20b and GPT-2 XL 1.5B.

Both gpt-oss and GPT-2 are decoder-only LLMs built on the transformer architecture introduced in the [Attention Is All You Need \(2017\)](#) paper. Over the years, many details have evolved.

However, these changes are not unique to gpt-oss. And as we will see later, they appear in many other LLMs. Since I discussed many of these aspects in the previous [Big Architecture Comparison](#) article, I will try to keep each subsection brief and focused.

2.1 Removing Dropout

[Dropout \(2012\)](#) is a traditional technique to prevent overfitting by randomly "dropping" (i.e., setting to zero) a fraction of the layer activations or attention scores (Figure 3) during training. However, dropout is rarely used in modern LLMs, and most models after GPT-2 have dropped it (no pun intended).

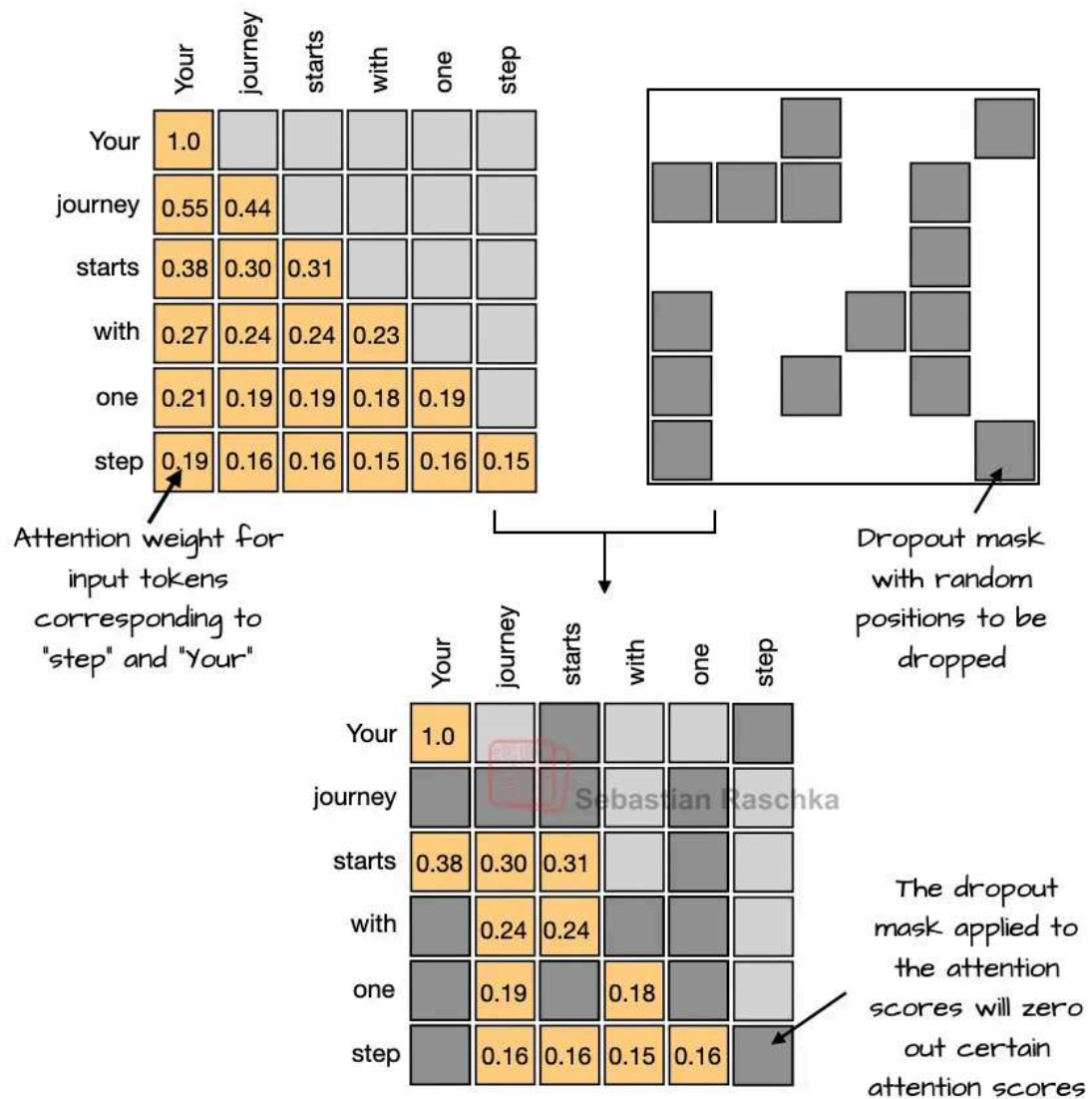


Figure 3: An illustration of dropout applied to the attention score matrix.

I assume that dropout was originally used in GPT-2 because it was inherited from the classic transformer architecture. Researchers likely noticed that it does not really improve LLM performance (I observed the same in my small-scale GPT-2 replication runs). This is likely because LLMs are typically trained for only a single epoch over massive datasets, which contrast to the multi-hundred-epoch training regimes for which dropout was first introduced. So, since LLMs see each token only once during training, there is little risk of overfitting.

Interestingly, while Dropout is kind of ignored in LLM architecture design for many years, I found a [2025 research paper](#) with small scale LLM experiments (Pythia 1.4B) that concluded that Dropout results in worse downstream performance in these single-epoch regimes.

2.2 RoPE Replaces Absolute Positional Embeddings

In transformer-based LLMs, positional encoding is necessary because of the attention mechanism. By default, attention treats the input tokens as if they have no order. In the original GPT architecture, absolute positional embeddings addressed this by adding a learned embedding vector for each position in the sequence (Figure 4), which is then added to the token embeddings.

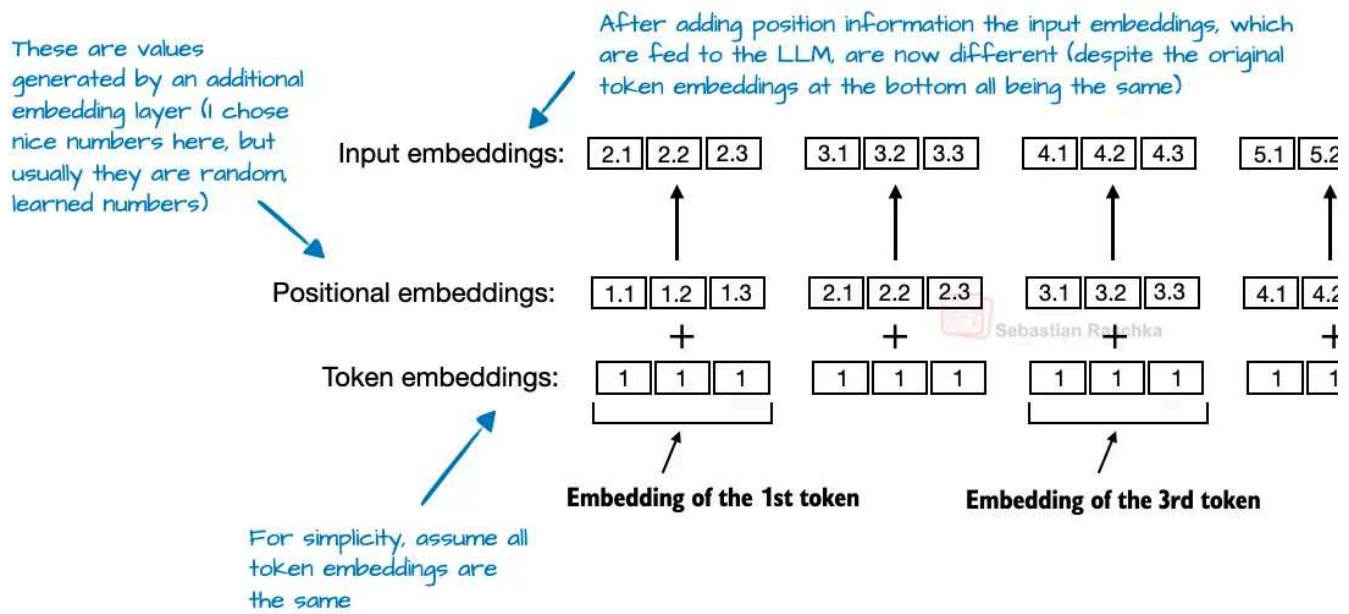


Figure 4: Illustration of absolute positional embeddings.

RoPE ([Rotary Position Embedding](#)) introduced a different approach: instead of adding position information as separate embeddings, it encodes position by rotating the query key vectors in a way that depends on each token's position. (RoPE is an elegant idea but a bit of a tricky topic to explain. I plan to cover separately in more detail one day.)

While first introduced in 2021, RoPE became widely adopted with the release of the original Llama model in 2023 and has since become a staple in modern LLMs.

2.3 Swish/SwiGLU Replaces GELU

Early GPT architectures used GELU. Why now use Swish over GELU? Swish (also referred as sigmoid linear unit or SiLU) is considered computationally slightly cheaper, and in my opinion, that's all there is to it. Depending on which paper you look at, you will find that it's slightly better than the other in terms of modeling performance. In my opinion, these sr

differences are probably within a standard error, and your mileage will vary based on hyperparameter sensitivity.

Activation functions used to be a hot topic of debate until the deep learning community largely settled on ReLU more than a decade ago. Since then, researchers have proposed many ReLU-like variants with smoother curves, and GELU and Swish (Figure 5) are ones that stuck.

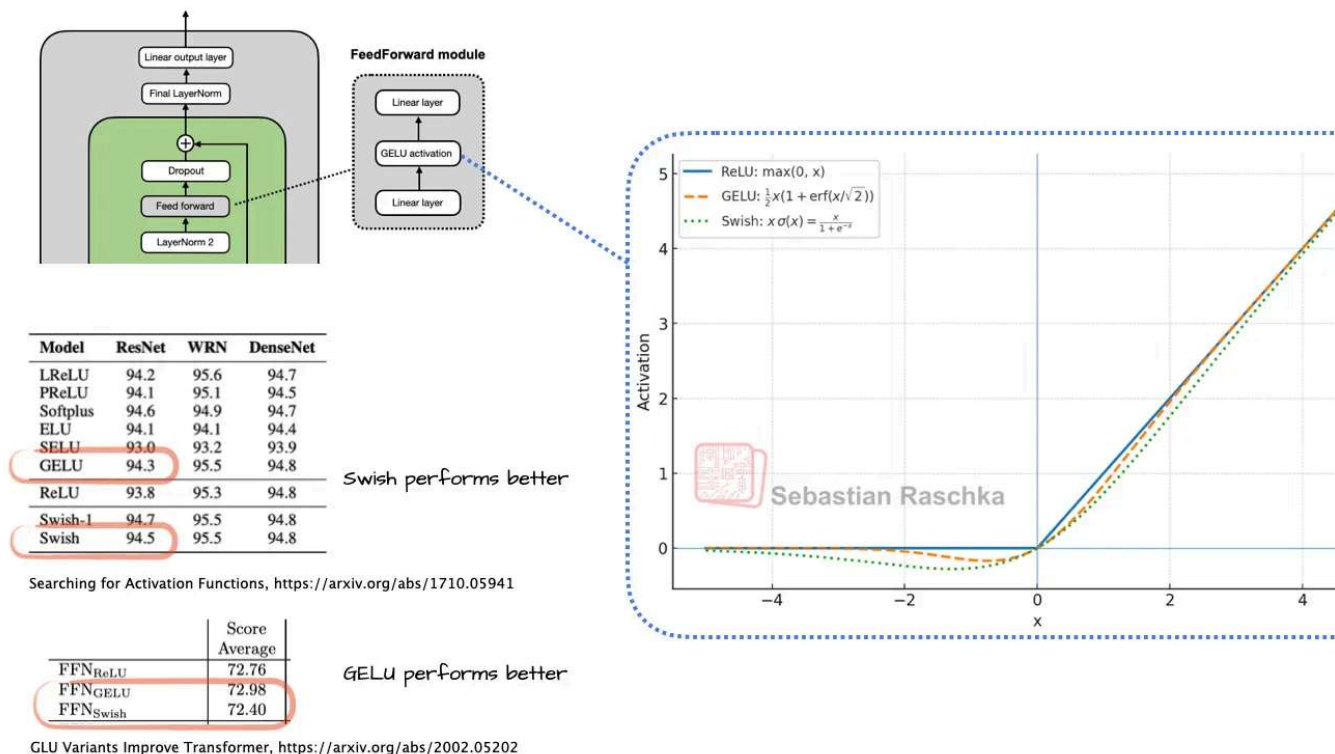


Figure 5: Comparison between Swish and GELU activations, which are both smoother versions of ReLU.

Early GPT architectures used GELU, which is defined as $0.5x * [1 + \text{erf}(x / \sqrt{2})]$. Here, erf (short for error function) is the integral of a Gaussian and it is computed using polynomial approximations of the Gaussian integral, which makes it more computationally expensive than simpler functions like the sigmoid used in Swish, where Swish is simply $\text{sigmoid}(x)$.

In practice, Swish is computationally slightly cheaper than GELU, and that's probably the main reason it replaced GELU in most newer models. Depending on which paper we look at,

one might be somewhat better in terms of modeling performance. But I'd say these gains are often within standard error, and the winner will depend heavily on hyperparameter tuning.

Swish is used in most architectures today. However, GELU is not entirely forgotten; for example, Google's Gemma models still use GELU.

What's more notable, though, is that the feed forward module (a small multi-layer perceptron) is replaced by a gated "GLU" counterpart, where GLU stands for gated linear unit and was proposed in a [2020 paper](#). Concretely, the 2 fully connected layers are replaced by 3 fully connected layers that are used as shown in Figure 6 below.

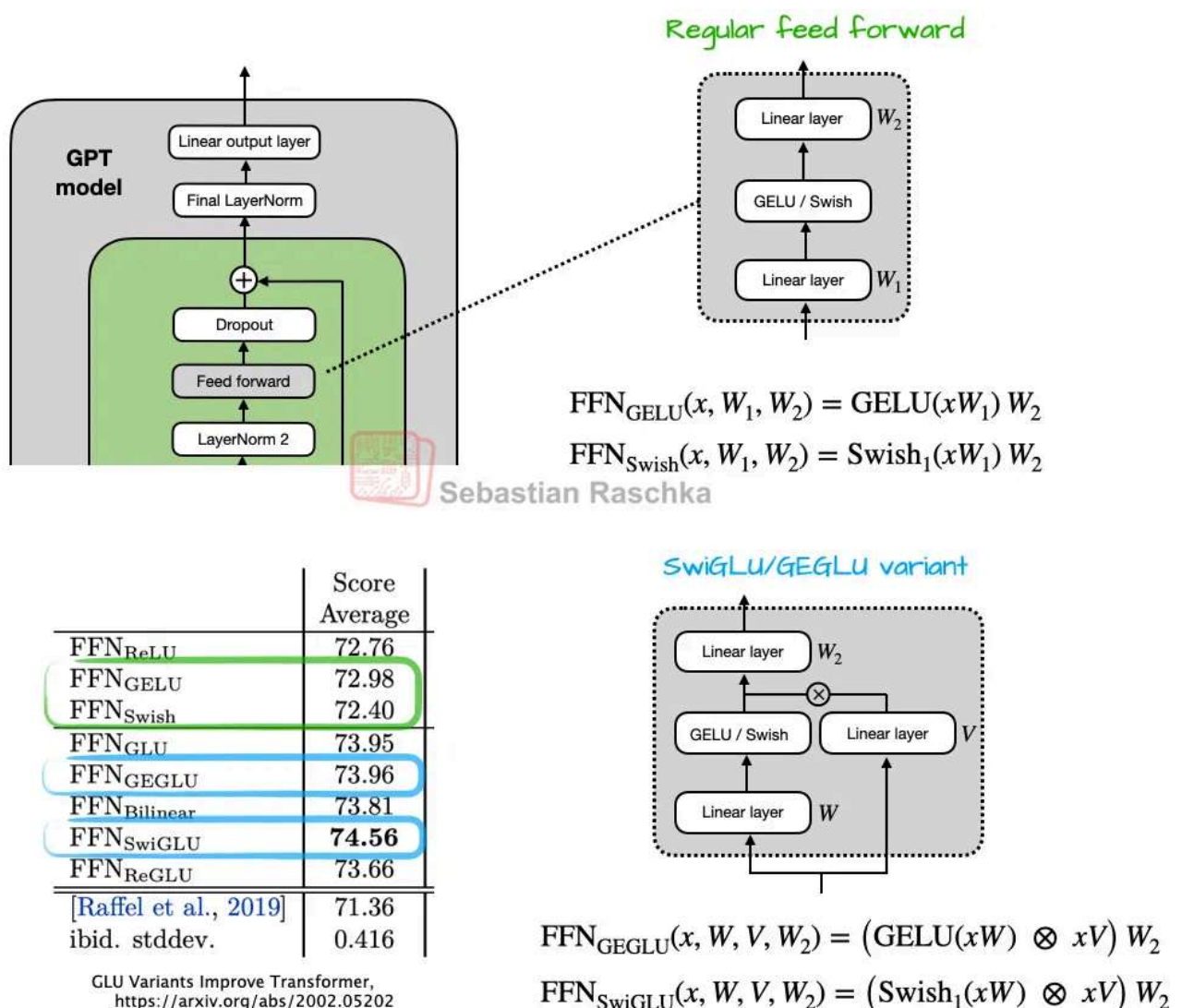


Figure 6: A comparison between Swish and GELU and their gated counterparts, SwiGLU and GEGLU.

At first glance, it may appear that the GEGLU/SwiGLU variants may be better than the feed forward layers because there are simply more parameters due to the extra layer. But is deceiving because in practice, the W and V weight layers in SwiGLU/GEGLU are usually chosen to be half the size each of the W_1 layer in a traditional feed forward layer.

To illustrate this better, consider the concrete code implementations of the regular and variants:

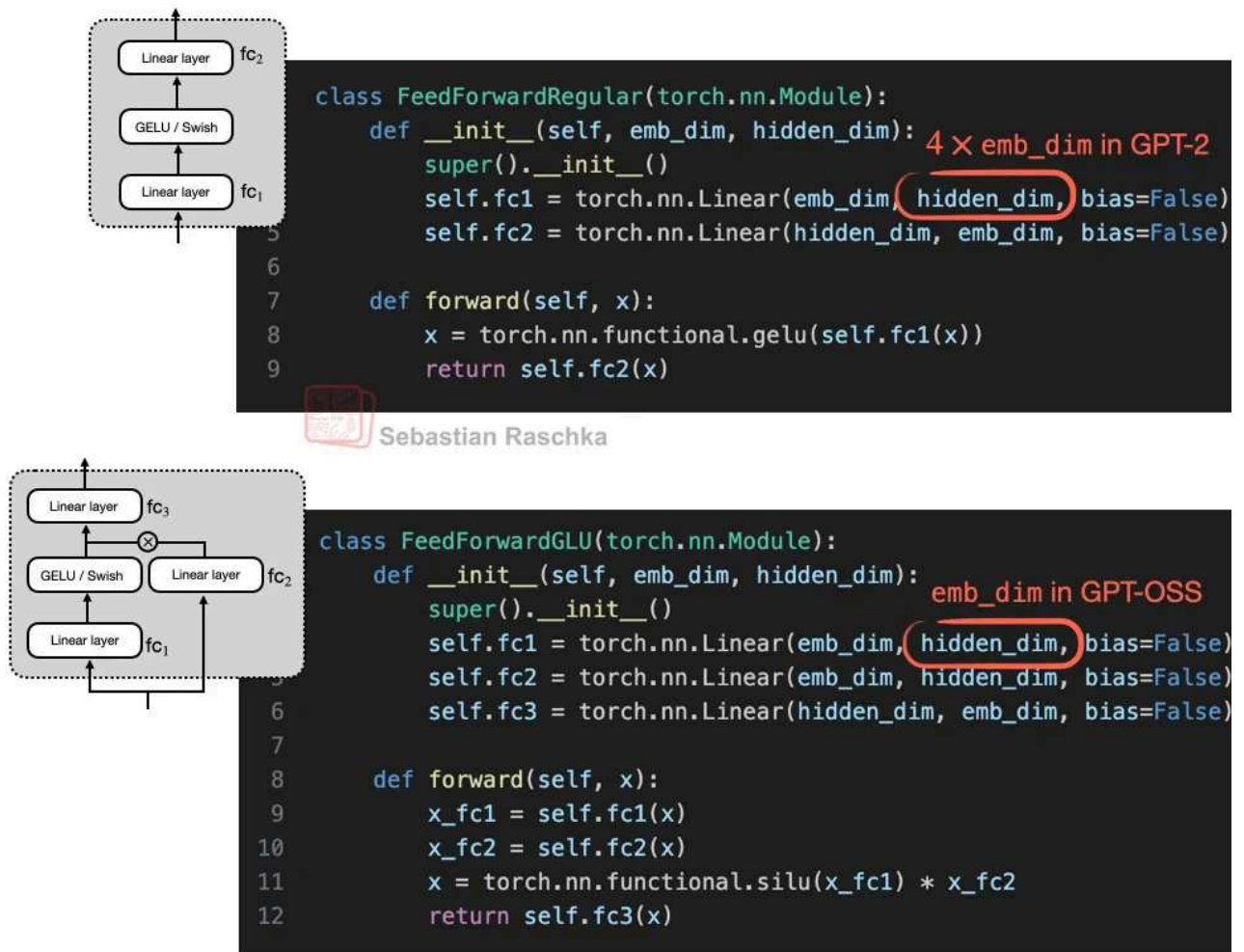


Figure 7: Regular feed forward module (top) and SwiGLU variant (bottom) next to each other. Note that the Swish function is implemented as "silu" in PyTorch.

So, suppose we have an embedding dimension of 1024. In the regular feed forward case, it would then be

- fc1: $1024 \times 4096 = 4,194,304$
- fc2: $1024 \times 4096 = 4,194,304$

That is $fc1 + fc2 = 8,388,608$ parameters.

For the GLU variant, we have

- $fc1: 1024 \times 1024 = 1,048,576$
- $fc2: 1024 \times 1024 = 1,048,576$
- $fc3: 1024 \times 1024 = 1,048,576$

I.e., $3 \times 1,048,576 = 3,145,728$ weight parameters.

So, overall, using the GLU variants results in fewer parameters, and they perform better well. The reason for this better performance is that these GLU variants provide an additional multiplicative interaction, which improves expressivity (the same reason deep & slim networks perform better than shallow & wide neural networks, provided they are trained well).

2.4 Mixture-of-Experts Replaces Single FeedForward Module

In addition to upgrading the feed forward module to a SwiGLU, as discussed in the previous section, gpt-oss replaces the single feed forward module with multiple feed forward modules using only a subset for each token generation step. This approach is known as a Mixture of Experts (MoE) and illustrated in Figure 8 below.

GPT-OSS 20B

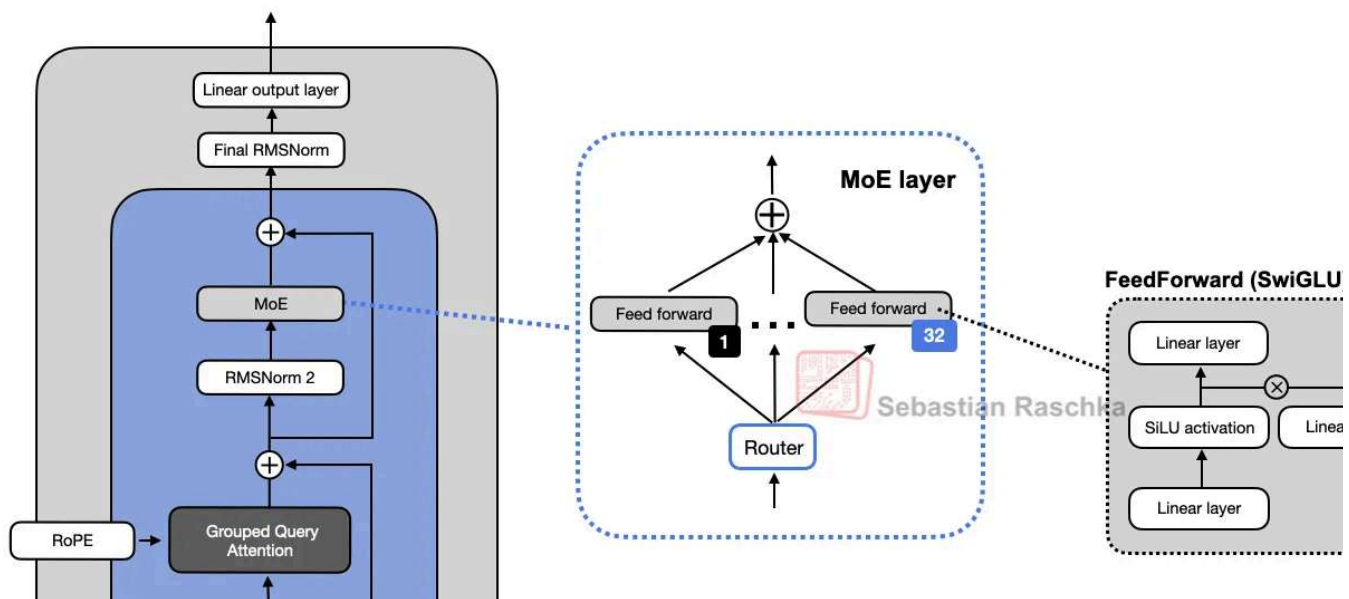


Figure 8: The feed forward module is replaced by a Mixture-of-Expert (MoE) module.

So, replacing *a single* feed forward module with *multiple* feed forward modules (as done in the MoE setup) substantially increases the model's total parameter count. However, the key is that we don't use ("activate") all experts for every token. Instead, a router selects only a small subset of experts per token.

Because only a few experts are active at a time, MoE modules are often referred to as *sparse* in contrast to *dense* modules that always use the full parameter set. However, the large number of parameters via an MoE increases the capacity of the LLM, which means it can take up more knowledge during training. The sparsity keeps inference efficient, though we don't use all the parameters at the same time.

(Fun fact: In most MoE models, expert weights account for more than 90% of the total parameters.)

2.5 Grouped Query Attention Replaces Multi-Head Attention

As mentioned in my previous articles, Grouped Query Attention (GQA) has emerged in recent years as a more compute- and parameter-efficient alternative to Multi-Head Attention (MHA).

In MHA, each head has its own set of keys and values. GQA reduces memory usage by grouping multiple heads to share the same key and value projections.

For example, as shown in Figure 9, if there are 2 key–value groups and 4 attention heads 1 and 2 might share one set of keys and values, while heads 3 and 4 share another. This grouping decreases the total number of key and value computations, leading to lower memory usage and improved efficiency without noticeably affecting modeling performance according to ablation studies.

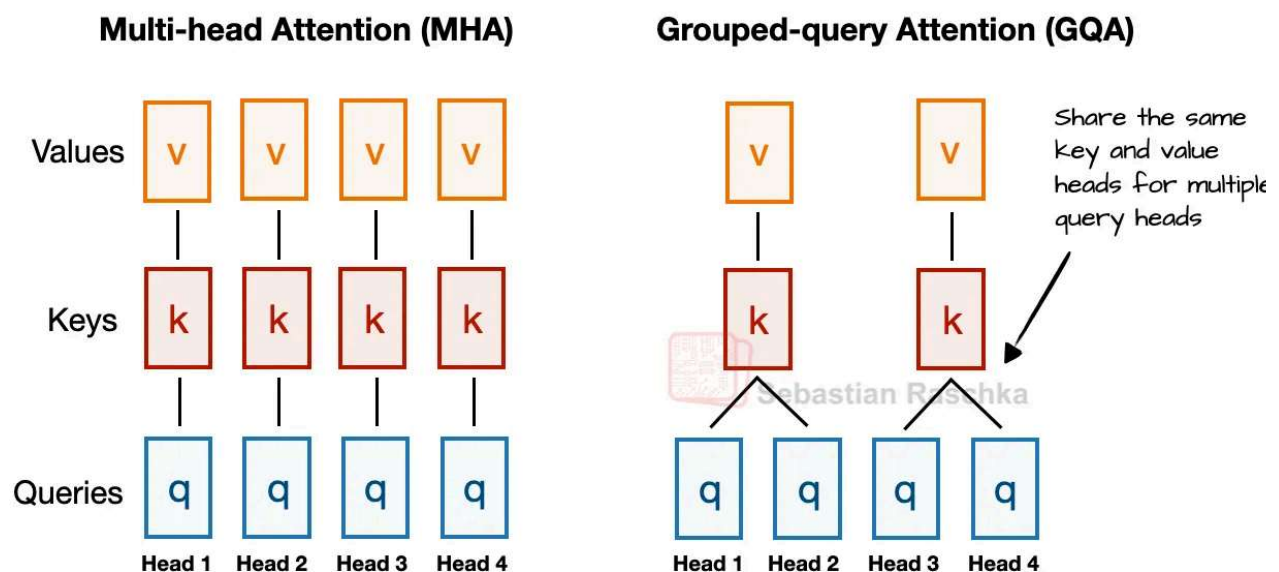


Figure 9: A comparison between MHA and GQA. Here, the group size is 2, where a key and value pair is shared among 2 queries.

So, the core idea behind GQA is to reduce the number of key and value heads by sharing them across multiple query heads. This (1) lowers the model's parameter count and (2) reduces the memory bandwidth usage for key and value tensors during inference since keys and values need to be stored and retrieved from the KV cache.

(If you are curious how GQA looks in code, see my [GPT-2 to Llama 3 conversion guide](#) version without KV cache and my KV-cache variant [here](#).)

While GQA is mainly a computational-efficiency workaround for MHA, ablation studies as those in the [original GQA paper](#) and the [Llama 2 paper](#)) show it performs comparably to standard MHA in terms of LLM modeling performance.

2.6 Sliding Window Attention

Sliding-window attention (Figure 10 below) was first introduced in the [LongFormer paper \(2020\)](#) and later popularized by Mistral. Interestingly, gpt-oss applies it in every second layer. You can think of it as a variation of multi-head attention, or in this case grouped query attention (GQA), where the attention context is restricted to a smaller window, reducing memory usage and compute costs.

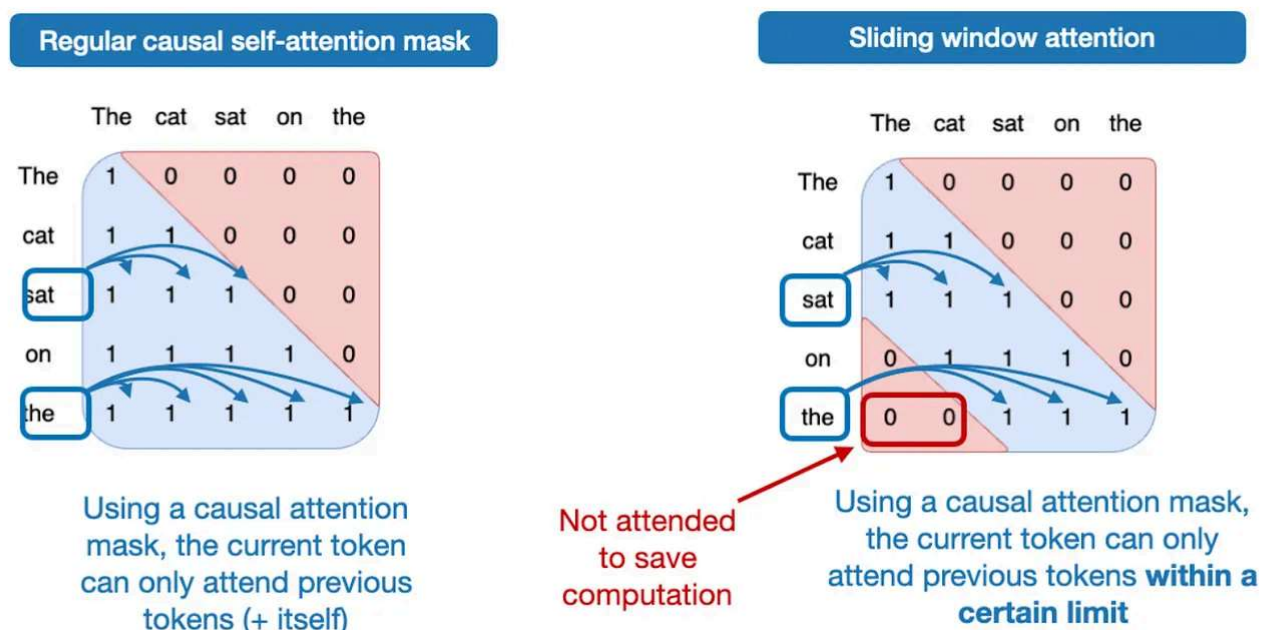


Figure 10: Comparison between regular attention (left) and sliding window attention (right).

Concretely, gpt-oss alternates between GQA layers that attend to the full context and layers with a sliding window limited to 128 tokens.

As I discussed in my [previous article](#), [Gemma 2 \(2024\)](#) used a similar 1:1 ratio. [Gemma](#) earlier this year went much further and shifted to a 5:1 ratio, which means only one full-attention layer for every five sliding-window (local) attention layers.

According to the Gemma ablation studies, sliding-window attention has minimal impact on modeling performance, as shown in the figure below. Note that the window size in Gemma 2 was 4096 tokens, which Gemma 3 reduced to 1024. In gpt-oss, the window is just 128 tokens, which is remarkably small.

And as a fun fact, the [official announcement article](#) notes that sliding-window attention apparently already used in GPT-3:

The models use alternating dense and locally banded sparse attention patterns, similar to GPT-3

Who knew!? I went back to the original [GPT-3 paper](#), and it was indeed mentioned the

We use the same model and architecture as GPT-2 [RWC+19], including the modified initialization, pre-normalization, and reversible tokenization described therein, with the exception that we use alternating dense and locally banded sparse attention patterns in the layers of the transformer, similar to the Sparse Transformer [CGRS19].

2.7 RMSNorm Replaces LayerNorm

Finally, the last small tweak, coming from GPT-2, is replacing [LayerNorm \(2016\)](#) by [RMSNorm \(2019\)](#), which has been a common trend in recent years.

Akin to swapping GELU with Swish and SwiGLU, RMSNorm is one of these smaller but sensible efficiency improvements. RMSNorm is similar to LayerNorm in its purpose to normalize layer activations, as shown in Figure 11 below.

You might recall that not too long ago, BatchNorm was the go-to choice for this task. It has since fallen out of favor, largely because it is harder to parallelize efficiently (due to the need for global variance batch statistics) and performs poorly with small batch sizes.

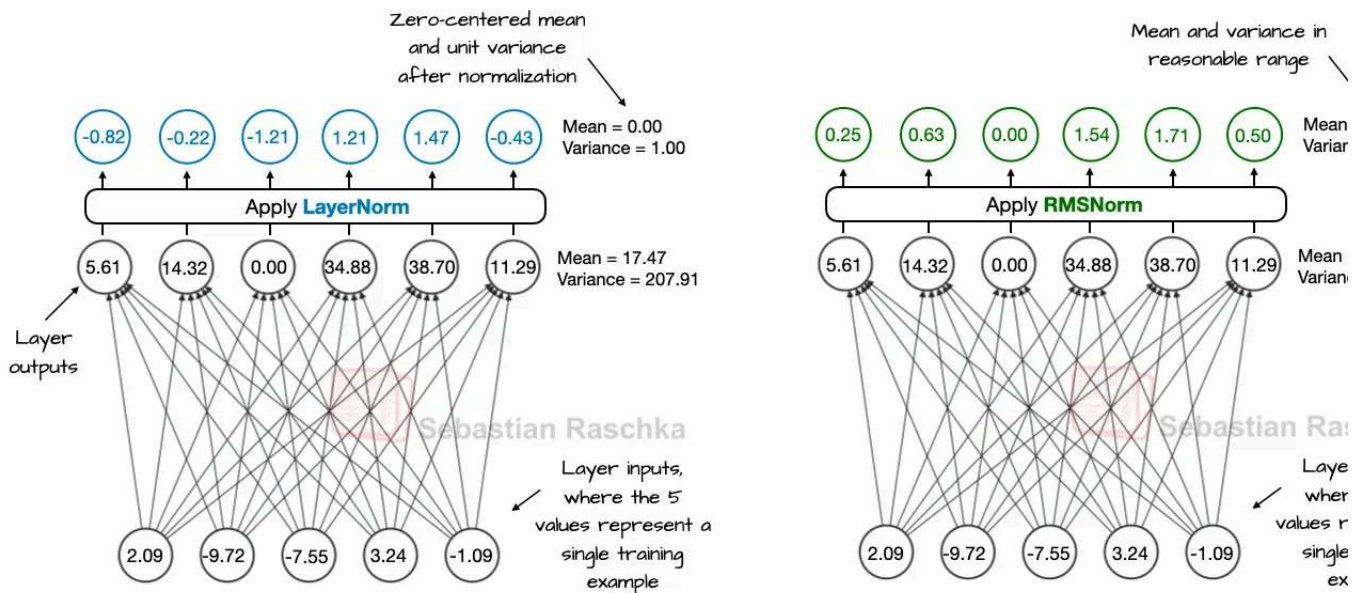


Figure 11: A comparison between LayerNorm (left) and RMSNorm (right) for a small linear layer.

As we can see in Figure 11 above, both LayerNorm and RMSNorm scale the layer outputs to be in a reasonable range.

LayerNorm subtracts the mean and divides by the standard deviation such that the layer outputs have a zero mean and unit variance (variance of 1 and standard deviation of 1).

RMSNorm divides the inputs by the root-mean-square. This scales activations to a comparable magnitude without enforcing zero mean or unit variance. In this particular example shown in Figure 11, the mean is 0.77 and the variance is 0.41.

Both LayerNorm and RMSNorm stabilize activation scales and improve optimization, but RMSNorm is often preferred in large-scale LLMs because it is cheaper to compute. Unlike LayerNorm, RMSNorm has no bias (shift) term and reduces the expensive mean and variance computations to a single root-mean-square operation. This reduces the number of cross-feature reductions from two to one, which lowers communication overhead on GPUs and improves training efficiency.

Figure 12 shows what this looks like in code:

```

1 class LayerNorm(torch.nn.Module):
2     def __init__(self, emb_dim):
3         super().__init__()
4         self.eps = 1e-5
5         self.scale = torch.nn.Parameter(torch.ones(emb_dim))
6         self.shift = torch.nn.Parameter(torch.zeros(emb_dim))
7
8     def forward(self, x):
9         mean = x.mean(dim=-1, keepdim=True)
10        var = x.var(dim=-1, keepdim=True, unbiased=False)
11        norm_x = (x - mean) / torch.sqrt(var + self.eps)
12        return self.scale * norm_x + self.shift

```

```

1 class RMSNorm(torch.nn.Module):
2     def __init__(self, emb_dim):
3         super().__init__()
4         self.eps = 1e-5
5         self.scale = torch.nn.Parameter(torch.ones(emb_dim))
6
7     def forward(self, x):
8         rms = torch.sqrt(torch.mean(x ** 2, dim=-1, keepdim=True) + self.eps)
9         norm_x = x / rms
10        return self.scale * norm_x

```

Figure 12: Code implementations of LayerNorm and RMSNorm showing that RMSNorm is computationally simpler.

2.8 The GPT-2 Legacy

I still think that GPT-2 is an excellent beginner architecture when learning about LLMs. simple enough to understand without getting lost in layers of optimization tricks, but st complex enough to give you a solid grasp of how modern transformer models work.

By starting with GPT-2, you can focus on the fundamentals (attention mechanisms, positional embeddings, normalization, and the overall training pipeline) without being overwhelmed by the extra features and tweaks found in newer architectures.

In fact, I think it's worth the time to learn about and even implement GPT-2 first before to stack newer changes on top. You will not only have an easier time understanding the changes, but you will likely also appreciate them more, because you will get a better understanding of what limitations or problems they try to solve.

For instance, starting with my GPT-2 code I recently implemented the [Qwen3 architec from scratch](#), which is super similar to gpt-oss, which brings us to the next topic: Comp

gpt-oss to a more recent architecture.

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3. Comparing gpt-oss To A Recent Architecture (Qwen3)

Now that we have walked through the evolution from GPT-2 to GPT OSS, we can take next step and compare GPT OSS to a more recent architecture, Qwen3, which was released three months earlier in May 2025.

The reason I am selecting Qwen3 here is that it is among the top open-weight models the time of writing. Additionally, one of the Qwen3 MoE models is more or less directly comparable to GPT OSS due to its relatively similar overall size in terms of trainable parameters.

Figure 13 below compares gpt-oss-20b to a Qwen3 model of comparable size.

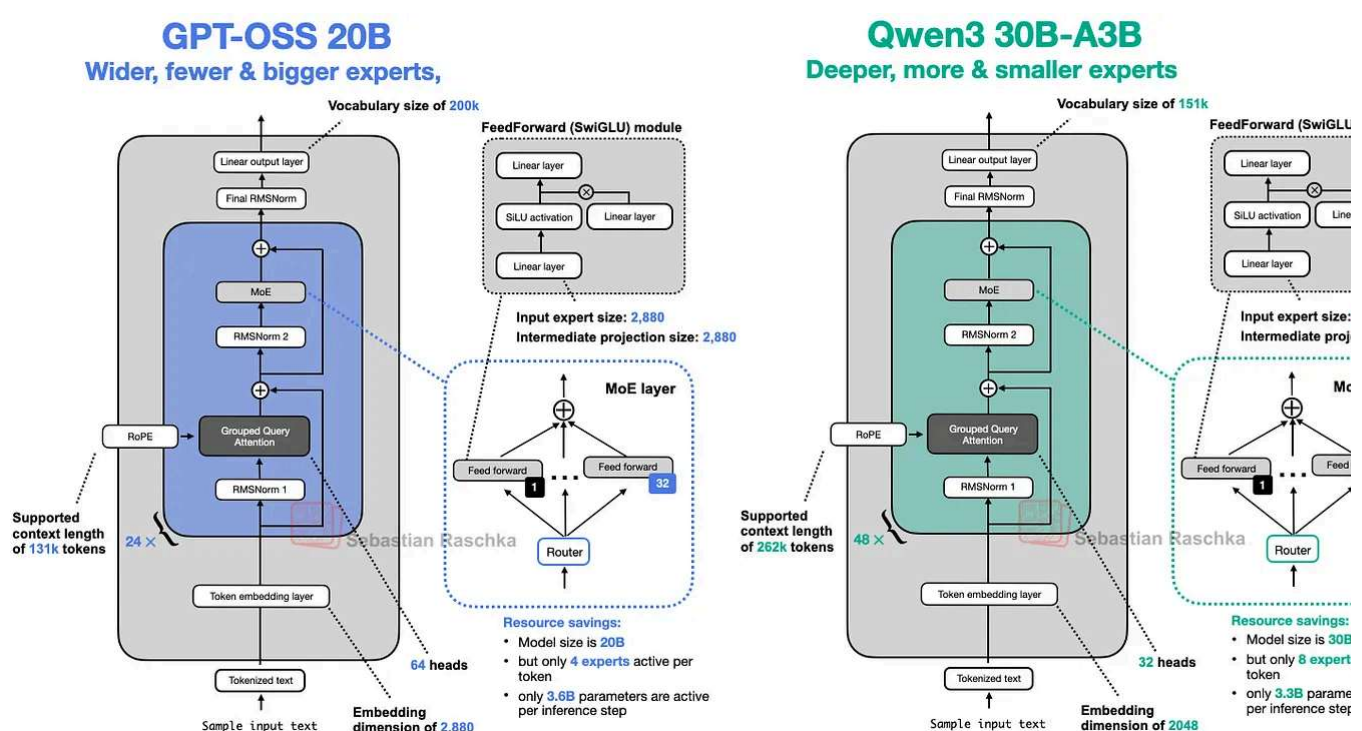


Figure 13: A gpt-oss and Qwen3 model of comparable size side by side.

As we can see, gpt-oss 20B and Qwen3 30B-A3B are very similar in their architecture components. The primary difference here, aside from the dimensions, is that gpt-oss employs sliding window attention, as discussed earlier in section 1.6 (not shown in this figure), whereas Qwen3 does not.

Let's walk through the noteworthy details one by one in the following subsections.

3.1 Width Versus Depth

If we look at the two models closely, we see that Qwen3 is a much deeper architecture with its 48 transformer blocks instead of 24 (Figure 14).

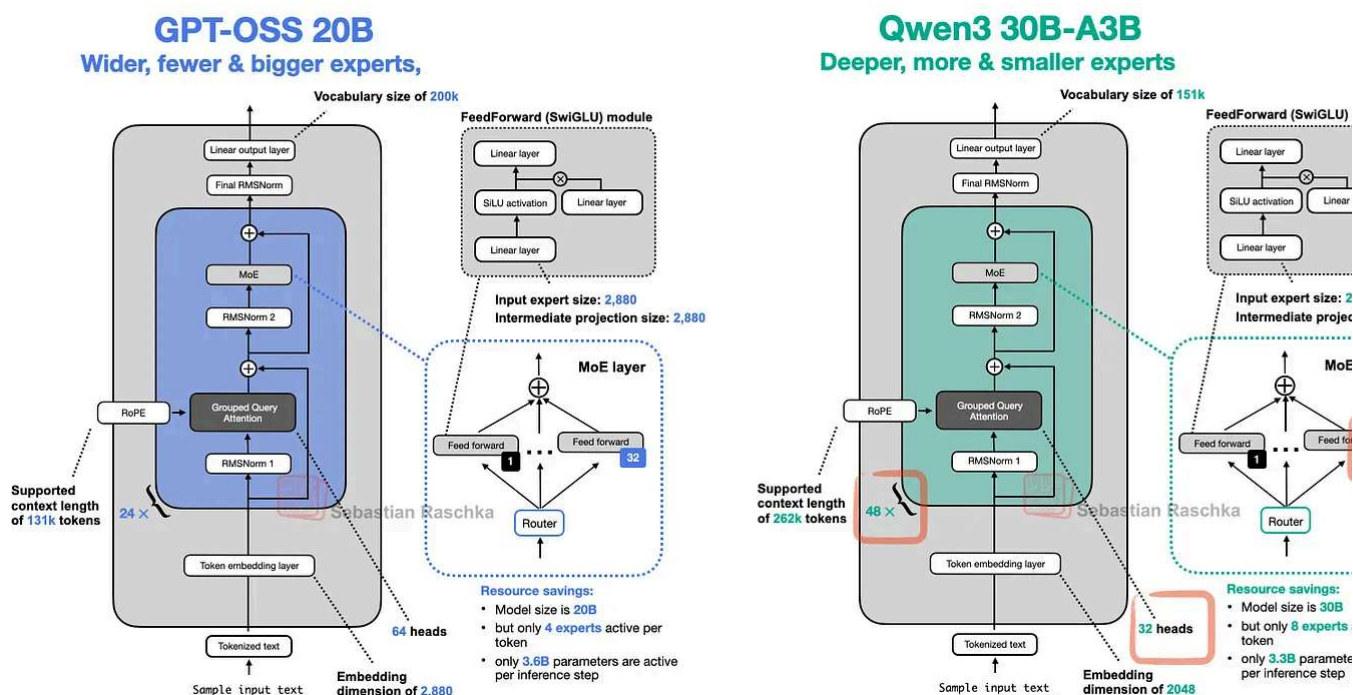


Figure 14: Qwen3 has twice as many transformer blocks as gpt-oss-20b.

On the other hand, gpt-oss is a much wider architecture:

- An embedding dimension of 2880 instead of 2048
- An intermediate expert (feed forward) projection dimension of also 2880 instead of 2048

It's also worth noting that gpt-oss uses twice as many attention heads, but this doesn't directly increase the model's width. The width is determined by the embedding dimension.

Does one approach offer advantages over the other given a fixed number of parameters? As a rule of thumb, deeper models have more flexibility but can be harder to train due to instability issues, due to exploding and vanishing gradients (which RMSNorm and short-cut connections aim to mitigate).

Wider architectures have the advantage of being faster during inference (with a higher tokens/second throughput) due to better parallelization at a higher memory cost.

When it comes to modeling performance, there's unfortunately no good apples-to-apples comparison I am aware of (where parameter size and datasets are kept constant) except an ablation study in the [Gemma 2 paper \(Table 9\)](#), which found that for a 9B parameter architecture, a wider setup is slightly better than a deeper setup. Across 4 benchmarks the wider model achieved a 52.0 average score, and the deeper model achieved a 50.8 average score.

3.2 Few Large Versus Many Small Experts

As shown in Figure 14 above, it's also noteworthy that gpt-oss has a surprisingly small number of experts (32 instead of 128), and only uses 4 instead of 8 active experts per token. However, each expert is much larger than the experts in Qwen3.

This is interesting because the recent trends and developments point towards more, smaller models as being beneficial. This change, at a constant total parameter size, is nicely illustrated in Figure 15 below from the DeepSeekMoE paper.

Early MoE: Has bigger and fewer experts, and activates only a few experts (here: 2)

Fine-grained MoE uses more but smaller experts, and activates more experts (here: 4)

MoE with shared expert: also uses many small experts, but adds a shared expert that is always active

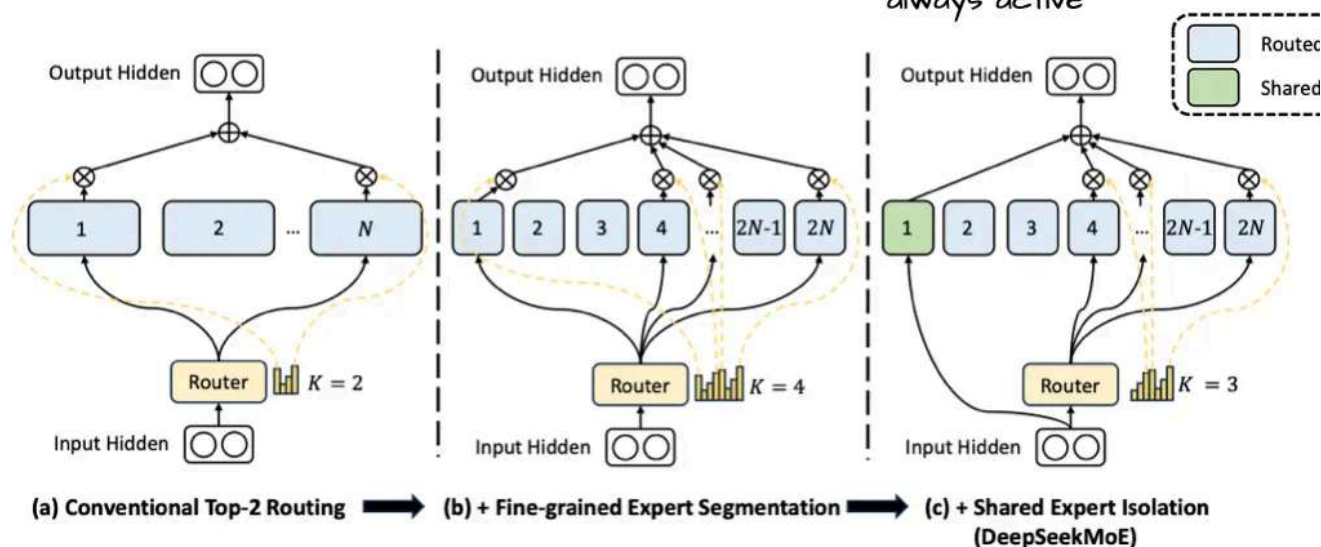


Figure 15: An annotated figure from "DeepSeekMoE: Towards Ultimate Expert Specialization in Mixture-of-Experts Language Models", <https://arxiv.org/abs/2401.06066>

Notably, unlike DeepSeek's models, neither gpt-oss nor Qwen3 uses shared experts, though.

To be fair, the small number of experts in gpt-oss could be a side effect of the 20B size. Looking at the 120B mode below, they indeed increased the number of experts (and transformer blocks) while keeping everything else fixed, as shown in Figure 16 below.

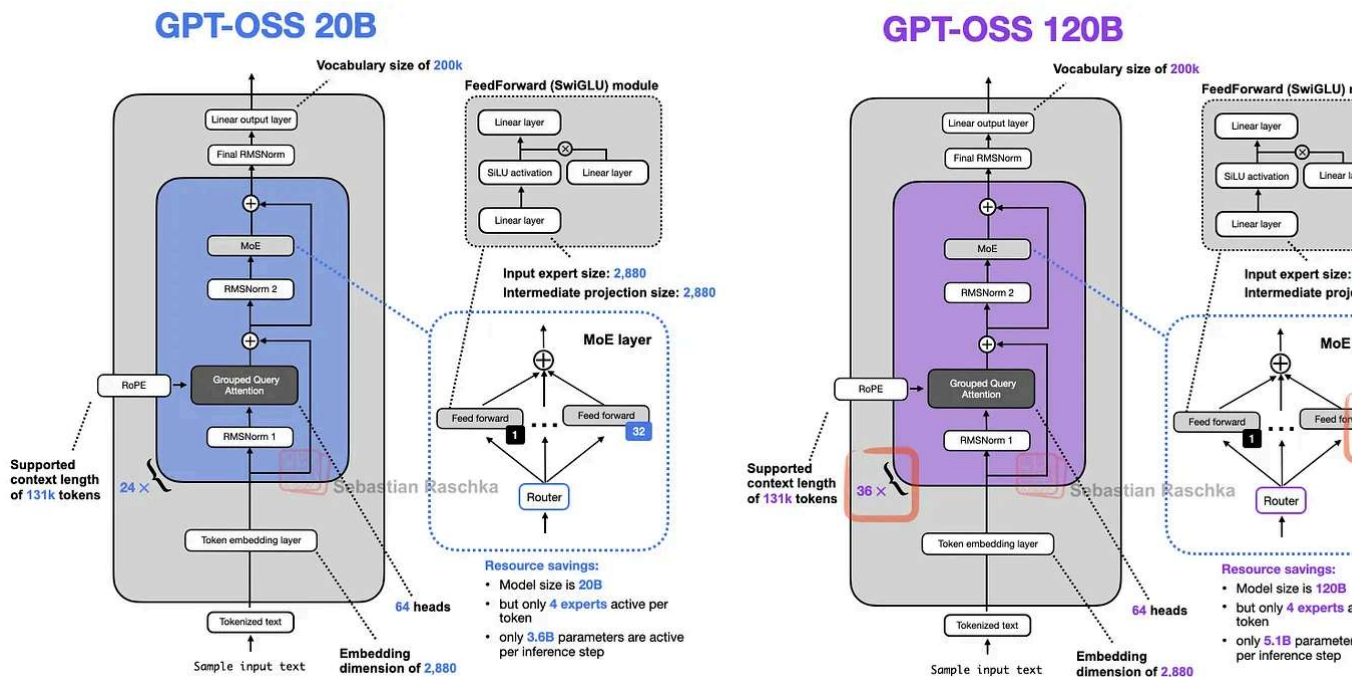


Figure 16: The two gpt-oss architectures side by side, where the larger 120B model only scales the number of transformer blocks and number of experts.

The boring explanation for the fact that the 20B and 120B models are so similar is probably that the 120B model was the main focus. And the easiest way to create a smaller model is to make it a bit shorter (fewer transformer blocks) and to reduce the number of experts because that's where most of the parameters are. However, one might speculate what they started training the 120B model, and then chopped some of the transformer block experts for continued pre-training (instead of starting from random weights).

In any case, it's because it's quite unusual to only scale those two (transformer blocks and number of experts). For instance, when looking at Qwen3 MoE models of multiple sizes (Figure 17 below), they were scaled more proportionally to each other over many more aspects..

Qwen3 Mixture-of-Experts

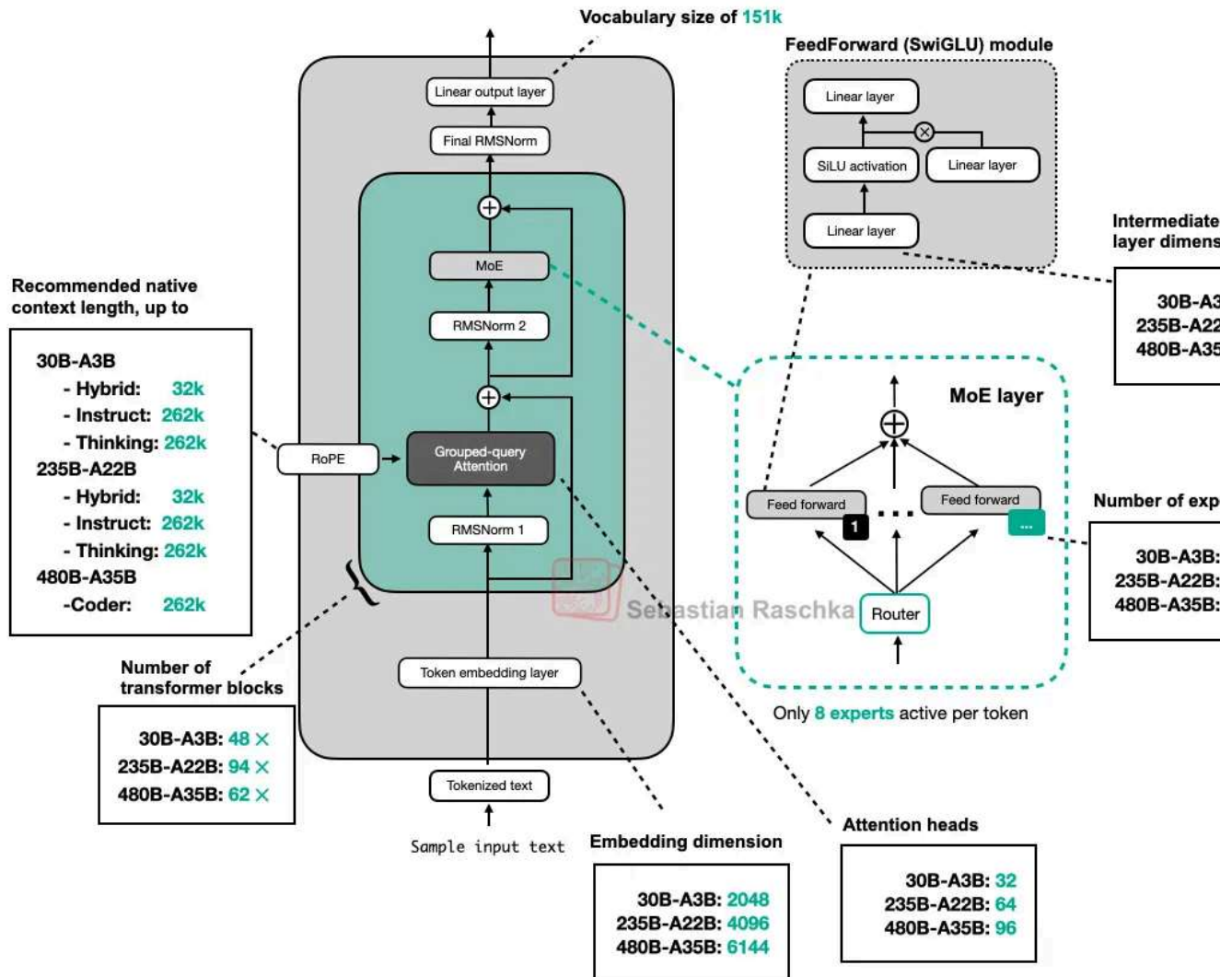


Figure 17: Architecture differences in the various Qwen3 models.

3.3 Attention Bias and Attention Sinks

Both gpt-oss and Qwen3 use grouped query attention. The main difference is that gpt restricts the context size via sliding window attention in each second layer, as mentioned earlier.

However, there's one interesting detail that caught my eye. It seems that gpt-oss uses units for the attention weights, as shown in the figure below.

```

228 class GptOssAttention(Qwen2Attention):
229     def __init__(self, config: GptOssConfig, layer_idx: int):
230         super().__init__(config, layer_idx)
231         self.q_proj = nn.Linear(
232             config.hidden_size, config.num_attention_heads * self.head_dim, bias=config.attention_bias
233         )
234         self.k_proj = nn.Linear(
235             config.hidden_size, config.num_key_value_heads * self.head_dim, bias=config.attention_bias
236         )
237         self.v_proj = nn.Linear(
238             config.hidden_size, config.num_key_value_heads * self.head_dim, bias=config.attention_bias
239         )
240         self.o_proj = nn.Linear(
241             config.num_attention_heads * self.head_dim, config.hidden_size, bias=config.attention_bias
242         )
243         self.sinks = nn.Parameter(torch.empty(config.num_attention_heads))

```

```

1 {
2   "architectures": [
3     "GptOssForCausalLM"
4   ],
5   "attention_bias": true,
6   "attention_dropout": 0.0,
7   "eos_token_id": 200002,
8   "experts_per_token": 4,
9   "head_dim": 64,
10  "hidden_act": "silu",
11  "hidden_size": 2880,
12  "initial_context_length": 4096,
13  "initializer_range": 0.02,
14  "intermediate_size": 2880,

```

Figure 18: gpt-oss models use bias units in the attention layers. See code example [here](#).

I haven't seen these bias units being used since the GPT-2 days, and they are commonly regarded as redundant. Indeed, I found a [recent paper](#) that shows mathematically that at least true for the key transformation (k_proj). Furthermore, the empirical results show there is little difference between with and without bias units (see Figure 19 below).

	GPT-2	RoBERTa-base
Original	2.9251	5.8890
No b_k	2.9250	5.8909

Test set loss; lower is better

Figure 19: Table from <https://arxiv.org/pdf/2302.08626> showing the average test loss when the models were trained from scratch with and without bias units.

Another detail you may have noticed is the definition of `sinks` in the code screenshot in Figure 18. In general models, attention sinks are special "always-attended" tokens placed at the start of the sequence to stabilize attention, which is especially useful in long-context scenarios. I.e., if the context gets very long, this special attended token at the beginning is always attended to, and it can learn to store some generally useful information about the entire sequence. (I think it was originally proposed in the [Efficient Streaming Language Models: Attention Sinks](#) paper.)

In the gpt-oss implementation, *attention sinks* are not actual tokens in the input sequence. Instead, they are learned per-head bias logits that are appended to the attention scores (Figure 20). The goal is the same as with the above-mentioned attention sinks, but without modifying the tokenized inputs.


```

228 class GptOssAttention(Qwen2Attention):
229     def __init__(self, config: GptOssConfig, layer_idx: int):
230         super().__init__(config, layer_idx)
231         self.q_proj = nn.Linear(
232             config.hidden_size, config.num_attention_heads * self.head_dim, bias=config.attention
233         )
234         self.k_proj = nn.Linear(
235             config.hidden_size, config.num_key_value_heads * self.head_dim, bias=config.attention
236         )
237         self.v_proj = nn.Linear(
238             config.hidden_size, config.num_key_value_heads * self.head_dim, bias=config.attention
239         )
240         self.o_proj = nn.Linear(
241             config.num_attention_heads * self.head_dim, config.hidden_size, bias=config.attention
242         )
243         self.sinks = nn.Parameter(torch.empty(config.num_attention_heads))
244
245     ...
246
247     key_states = repeat_kv(key, module.num_key_value_groups)
248     value_states = repeat_kv(value, module.num_key_value_groups)
249     attn_weights = torch.matmul(query, key_states.transpose(2, 3)) * scaling
250     if attention_mask is not None:
251         causal_mask = attention_mask[:, :, :, : key_states.shape[-2]]
252         attn_weights = attn_weights + causal_mask
253
254     sinks = module.sinks.reshape(1, -1, 1, 1).expand(query.shape[0], -1, query.shape[-2], -1)
255     combined_logits = torch.cat([attn_weights, sinks], dim=-1)

```

Figure 20: The use of attention sinks in gpt-oss; based on the Hugging Face code [here](#).

3.4 License

Lastly, and similar to Qwen3, the gpt-oss models are Apache 2.0 open-source license, is great (it's the same license that I prefer for my own open-source projects). This means the models can be distilled into other models or used in commercial products without restriction.

Open-weight vs. open-source LLMs. This distinction has been debated for years, but worth clarifying to avoid confusion about this release and its artifacts. Some model developers release only the model weights and inference code (for example, Llama, Gemma, gpt-oss), while others (for example, OLMo) release everything including training code, datasets, and weights as true open source.

By that stricter definition, gpt-oss is an *open-weight* model (just like Qwen3) because it includes the weights and inference code but not the training code or datasets. However,

terminology is used inconsistently across the industry.

I assume the "oss" in "gpt-oss" stands for *open source software*; however, I am positively surprised that OpenAI itself clearly describes gpt-oss as an open-weight model in their official [announcement article](#).

4 Other Interesting Tidbits

While the previous sections described how the architecture has evolved since GPT-2 and discussed its similarities to Qwen3 (and most other recent models), there are still a few additional but noteworthy details I have not mentioned, yet. These are points that did not fit neatly into the earlier sections but are still worth mentioning.

4.1 Training Overview

Unfortunately, there is not much information about the training set sizes and algorithms available. I added the most interesting puzzle pieces from the [model card report](#) (1) and [announcement post](#) (2) below:

The gpt-oss models were trained using our most advanced pre-training and post-training techniques [...] (1)

[...] required 2.1million H100-hours to complete, with gpt-oss-20b needing almost 10x fewer. (1)

[...] including a supervised fine-tuning stage and a high-compute RL stage [...] (2)

We trained the models on a mostly English, text-only dataset, with a focus on STEM, coding, and general knowledge. (2)

So, we know that the gpt-oss models are reasoning models. The training compute of 2 million H100 GPU hours is roughly on par with the 2.788 million H800 GPU hours that the ~5.6x larger [DeepSeek V3](#) model was trained for. Unfortunately, there is no information about the Qwen3 training time available yet.

Interestingly, the GPT-oss training hour estimate includes both the supervised learning instruction following and the reinforcement learning for reasoning, whereas DeepSeek just a pre-trained base model on top of which DeepSeek R1 was trained separately.

4.2 Reasoning Efforts

As mentioned in the previous section, the gpt-oss models are reasoning models. However what's particularly interesting is that they were trained so that users can easily control degree of reasoning via inference time scaling.

Concretely, gpt-oss models can receive "Reasoning effort: low/medium/high" instruction part of their system prompt, which directly affects the response length and accuracy, as shown in Figure 21.

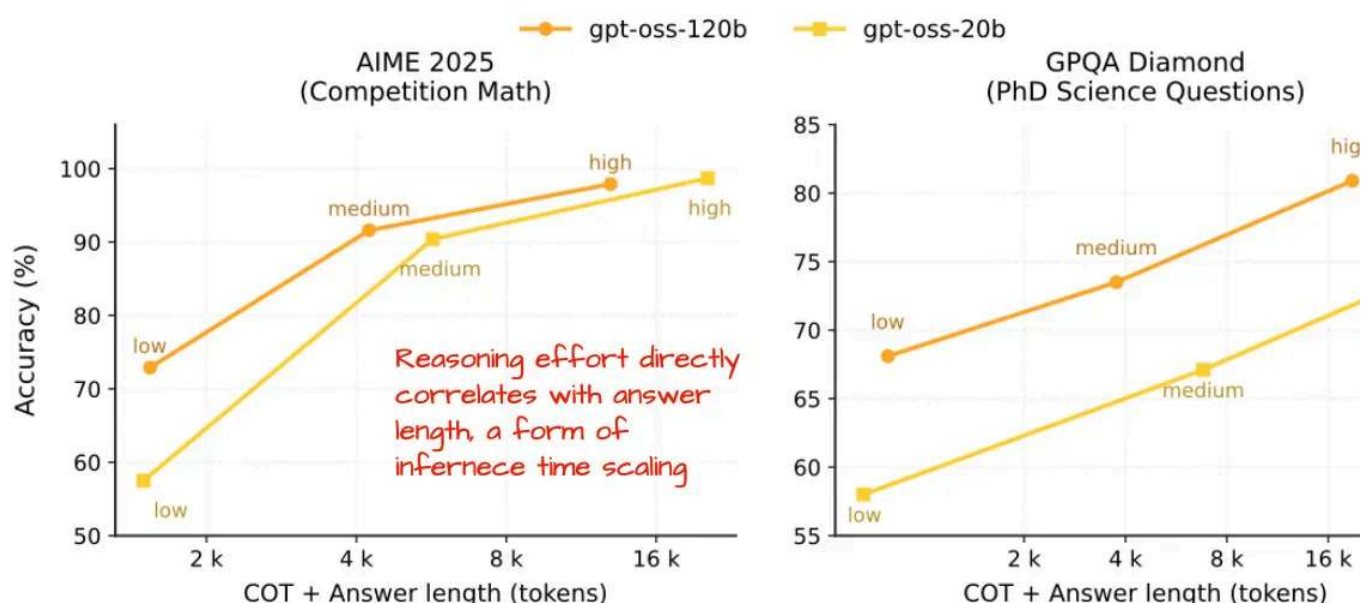


Figure 21: Response length and quality of gpt-oss models under different reasoning efforts
(annotated figure from the [model card](#))

This level of adjustability is useful because it lets us balance cost, compute, and accuracy. For example, if the task is simple, such as answering a straightforward knowledge question or fixing a small typo, we can skip extended reasoning. This saves time and resources while avoiding unnecessarily long responses and verbose reasoning traces.

It is somewhat unfortunate that OpenAI did not release the base models prior to reinforcement learning-based reasoning training, unlike Qwen3 or OLMo. Base models are particularly valuable starting points for researchers working on reasoning methods (with one reason I currently like working with Qwen3 Base). My guess is that OpenAI's decision was driven more by industry and production use cases than by research consideration.

Note that the original Qwen3 models also have a toggle for enabling/disabling thinking (reasoning) modes (via a `enable_thinking=True/False` setting in the tokenizer that adds `<think></think>` tags to disable the reasoning behavior). However, the Qwen3 team updated their models in the last few weeks and moved away from the hybrid model to dedicated Instruct/Thinking/Coder variants.

The reason was that the hybrid mode resulted in lower performance compared to the individual models:

After discussing with the community and reflecting on the matter, we have decided to abandon the hybrid thinking mode. We will now train the Instruct and Thinking models separately to achieve the best possible quality. [Source](#)

4.3 MXFP4 Optimization: A Small But Important Detail

One interesting surprise is that OpenAI released the gpt-oss models with an MXFP4 quantization scheme for the MoE experts.

Quantization formats used to be a niche topic, mostly relevant to mobile or embedded devices, but that's changed with the push toward bigger models. In this case, the MXFP4 optimization allows the model to run on single GPU devices.

Here's what that looks like in practice:

- The large model (think 120B) fits on a single 80GB H100 or newer GPU. Not considering hardware, but hey, it's much cheaper to rent a 1-H100 machine than a multi-H100 machine. Plus, we don't have to worry about distributing the model across GPUs and adding communication overhead. It's really nice that AMD MI300X cards are supported from day 1 as well!

- The smaller 20B model even fits into 16 GB of VRAM; the caveat is that it has to be RTX 50-series GPU or newer to support MXFP4. (Edit: support for older cards, such as RTX 4090, was recently added via a [patch](#).)

Note that the models will also run on older hardware but without MXFP4 support and will thus consume more RAM. Without MXFP4 optimization, the models in bfloat16 will consume more like 48 GB (gpt-oss-20b) and 240 GB (gpt-oss-120b).

By the way, I can run the gpt-oss-20b model comfortably on my Mac Mini using ollama; it uses about 13.5 Gb of memory, which is really reasonable.

4.4 Benchmarks

The models are still a bit too new for independent benchmarks. Checking the [LM Arena leaderboard](#), I found that gpt-oss is not listed, yet. So, Qwen3-Instruct remains the top weight model, according to users on the LM Arena, for now (Figure 22).

Rank (UB) ↑	Model ↓	Score ↓	95% CI (±) ↓	Votes ↓	Organization ↓	License ↓
1	 gpt-5	1481	±11	3,182	OpenAI	Proprietary
2	 gemini-2.5-pro	1460	±6	26,703	Google	Proprietary
2	 o3-2025-04-16	1450	±5	32,692	OpenAI	Proprietary
3	 chatgpt-4o-latest-20250326	1442	±5	31,219	OpenAI	Proprietary
4	 gpt-4.5-preview-2025-02-27	1438	±6	15,271	OpenAI	Proprietary
5	 grok-4-0709	1429	±6	13,314	xAI	Proprietary
5	 qwen3-235b-a22b-instruct-2507	1428	±9	4,831	Alibaba	Apache 2.0
6	 claude-opus-4-20250514-thinking-16k	1420	±6	18,461	Anthropic	Proprietary
6	 kimi-k2-0711-preview	1420	±6	12,400	Moonshot	Modified MIT
6	 glm-4.5	1416	±9	4,872	Z.ai	MIT
7	 deepseek-r1-0528	1417	±6	18,662	DeepSeek	MIT
8	 claude-opus-4-20250514	1412	±5	26,605	Anthropic	Proprietary
10	 grok-3-preview-02-24	1409	±5	32,182	xAI	Proprietary
10	 gemini-2.5-flash	1408	±5	31,987	Google	Proprietary
10	 gpt-4.1-2025-04-14	1407	±5	25,278	OpenAI	Proprietary
11	 qwen3-235b-a22b-thinking-2507	1398	±9	4,302	Alibaba	Apache 2.0

Figure 22: Current view of the LM Arena Leaderboard (as of 8 Aug 2025)

Looking at a reasoning benchmarks provide in the gpt-oss announcement post, we can see that the gpt-oss models are on par with OpenAI's proprietary models as well as Qwen3 (Figure 23).

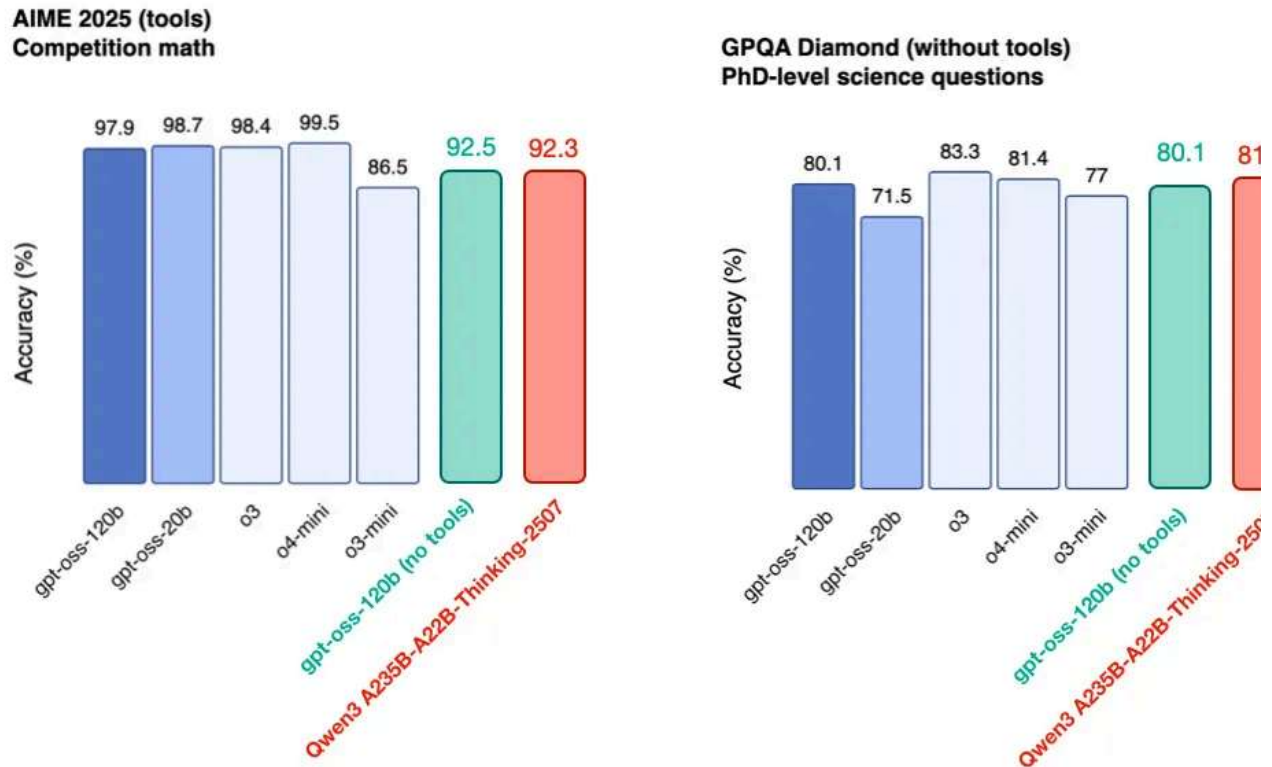


Figure 23: The main benchmark charts are from the official [gpt-oss announcement post](#).

The "no tools" gpt-oss-120b data is taken from the official [model card paper](#), and the

Qwen3 numbers are taken from the official [Qwen3 repository](#).

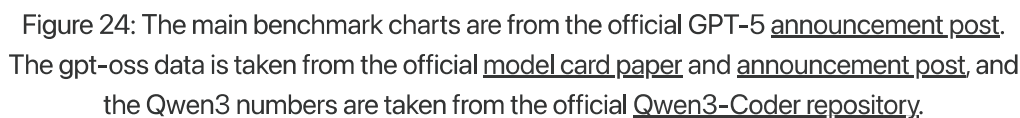
However, this should be caveated by the fact that gpt-oss-120b is almost half the size Qwen3 A235B-A22B-Thinking-2507 model and can run on a single GPU.

Benchmark performance, however, does not always reflect real-world usability. In my li use over the past few days, I have found gpt-oss to be quite capable. That said, as othe have observed, it does seem to have a relatively high tendency to hallucinate (a point a mentioned in its model card).

This may stem from its heavy training focus on reasoning tasks such as math, puzzles, code, which could have led to some "general knowledge forgetting." Still, because gpt- was designed with tool use in mind, this limitation may become less relevant over time. integration in open-source LLMs is still in its early stages, but as it matures, I expect the increasingly let models consult external sources (like search engines) when answering factual or knowledge-based queries.

If that happens, it could be sensible to prioritize reasoning capacity over memorization. is much like in human learning in school (or in life in general), where problem-solving st

5 gpt-oss and GPT-5



All in all, even though some people called the release overhyped, I am glad that we have a new set of really strong open weight models that are not too far behind the best proprietary ones. Of course, benchmarks often do not accurately reflect real-world use, and it is still early to tell based on the limited usage. But I think these are good times for people who work with open-weight and local (or privately hosted) models.

This magazine is a personal passion project, and your support helps keep it alive. If you would like to contribute, there are a few great ways:

- [Grab a copy of my book](#). *Build a Large Language Model (From Scratch)* walks you through building an LLM step by step, from tokenizer to training.
- [Check out the video course](#). There's now a 17-hour video course based on the book available from Manning. It follows the book closely, section by section, and works both as a standalone or as a code-along resource. The video course is ad-free (the YouTube version) and has a cleaner, more structured format. It also contains additional hours of pre-requisite video material created by Abhinav Kimothi.
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Build an LLM from scratch

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Open "https://livevideo.manning.com/module/7820_2_5" in a new tab

video

search liveVideo...

ch02.ipynb

File Edit View Run Kernel Tabs Settings Help

Python 3 (pykernel)

ch02.ipynb

the-verdict.txt

25 ago

print(result)

['Hello', ' ', ' ', 'world', ' ', ' ', 'This', ' ', ' ', 'is', ' ', ' ', 'a', ' ', ' ', 'test']

result = re.split(r'([,])', text)

print(result)

['Hello', ' ', ' ', ' ', 'world', ' ', ' ', ' ', 'This', ' ', ' ', ' ', 'is', ' ', ' ', 'a', ' ', ' ', 'test', ' ', ' ', '']

Mode Command

11:04/14:10

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interactive transcript

So let's do that with this regular expression and then maybe print the result and see how it looks like. And yeah, we can see, we have now the individual words and we have the space characters and so forth. Um, something we might want to do is also have the punctuation as, um, separate characters. So for that, we would make a little bit of a more sophisticated regular expression.

And like I told you, I'm not very good at regular expressions. So let me just write one and paste it here. So this is now a regular expression that is slightly more sophisticated. Um, so this one will include the punctuation as separate tokens where before they were part of the word itself. Okay.

So this is our simplest, um, way of knowing, tokenizing. Now as you have seen here, if I go here, there are actually white space characters, um, in the output here. So one thing we, to make this could be, for example, that we

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<https://arxiv.org/abs/2506.21734>

Do you already have a say about it?

TY for all your work! 

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